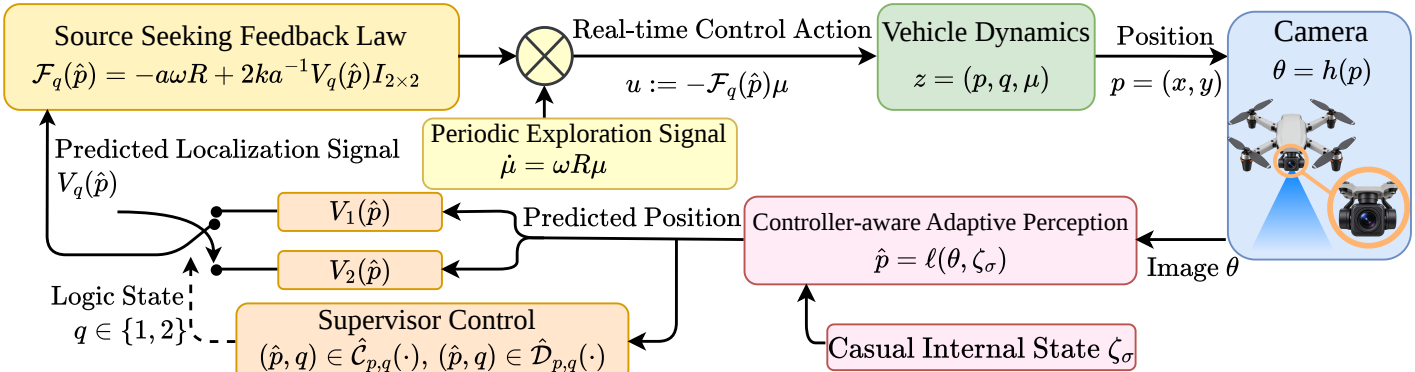

Source Seeking Feedback Law
$\mathcal{F}_q(\hat{p}) = -a\omega R + 2ka^{-1}V_q(\hat{p})I_{2\times 2}$
Real-time Control Action
$u := -\mathcal{F}_q(\hat{p})\mu$
Vehicle Dynamics
$z = (p, q, \mu)$
Position
$p = (x, y)$
Camera
$\theta = h(p)$
Periodic Exploration Signal
$\dot{\mu} = \omega R \mu$
Predicted Localization Signal
$V_q(\hat{p})$
$V_1(\hat{p})$
$V_2(\hat{p})$
Predicted Position
Controller-aware Adaptive Perception
$\hat{p} = \ell(\theta, \zeta_\sigma)$
Image θ
Logic State
$q \in \{1, 2\}$
Supervisor Control
$(\hat{p}, q) \in \hat{\mathcal{C}}_{p,q}(\cdot)$, $(\hat{p}, q) \in \hat{\mathcal{D}}_{p,q}(\cdot)$
Casual Internal State ζ_σ